



Multiple red blood cell flows through microvascular bifurcations: Cell free layer, cell trajectory, and hematocrit separation

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ABSTRACT

Multiple red blood cell (RBC) flows through a symmetric microvascular bifurcation model have been simulated with the two-dimensional immersed-boundary lattice-Boltzmann method. The cell free layer (CFL), the RBC separation process and trajectories, and the resulting hematocrit distributions in the daughter branches have been examined, and the effects of cell deformability, aggregation, and feeding hematocrit on the RBC separation have also been investigated. Our results show that the overall phase separation behavior is mainly related to the RBC distribution in the feeding flow (i.e., the CFL thickness). On the other hand, for individual RBCs, the hydrodynamic interaction plays a non-negligible role in determining their trajectories and destinations. A detailed examination of the flow and pressure fields in the bifurcation region indicates that the difference in flow pressure across the front and rear ends of a flowing RBC is the major driving force for the cell motion; while the shear stress on the back of a cell that has been pressed against the corner wall is responsible for the cell's slow sliding into a vessel branch. The results have also been compared with experimental studies, and reasonable agreement has been observed. The results and information from this study could be helpful for understanding the complex RBC separation process and its effects in microcirculation and relevant biomedical applications.

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Introduction

Continuously circulating through the body, blood performs many crucial biological functions, including supply of oxygen and nutrients to and removal of metabolic wastes from tissues, circulation of white blood cells, antibodies and platelets for immunization and self-repair, and regulation of body pH and temperature. These blood functions are mainly accomplished in the microvascular network composed of microvessels and capillaries. As observed in fundamental studies and clinical observations, abnormal microscopic blood flow behaviors are often associated with various diseases and disorders, such as heart diseases, hypertension, diabetes, cancers, malaria, anemia, and atherosclerosis (Popel and Johnson, 2005; Stoltz et al., 1999). For this reason, microcirculation has attracted broad interests and extensive studies have been conducted with the help of advances in experimental and computational techniques.

At a microscopic level, blood is a concentrated suspension of several cellular components, including red blood cells (RBCs) or erythrocytes, white blood cells or leucocytes, and platelets, suspended in plasma, an

aqueous solution of numerous ions and macromolecules. RBCs are the most important component of blood for their crucial biological functions as the oxygen carriers, and their direct effects on hemodynamics and hemorheology due to the large number density (~ 5 million/mm³). RBCs possess several particular cellular features, including a biconcave shape, a highly flexible membrane, and an anuclear fluidic interior (the cytoplasm). These features provide RBCs with a high deformability to travel through narrow passages even smaller than RBCs themselves. In addition, RBCs can aggregate and form one-dimensional stacks-of-coins-like rouleaux or three-dimensional (3D) aggregates (Baskurt et al., 2011; Popel and Johnson, 2005; Stoltz et al., 1999). Increased aggregation is often observed in several clinical states, such as diabetes, heart diseases, inflammation, acute myocardial infarction, and bacterial sepsis (Baskurt and Meiselman, 2007; Baskurt et al., 2011).

When flowing in microvessels, blood also exhibits unique flow structures and rheological behaviors, such as a cell free layer (CFL) near the vessel surface, a RBC-rich core in the center region, a flattened velocity profile, as well as the vessel-size-dependent effective viscosity (the so-called Fahraeus–Lindqvist effect) (Popel and Johnson, 2005). More interesting, at a microvascular bifurcation, the daughter vessel with a higher flow rate will receive an elevated hematocrit (volume fraction occupied by RBCs in blood) compared with the mother vessel (Pries et al., 1986, 1996). This phenomenon, usually called the hematocrit phase separation, is mainly from the particular RBC distribution in the mother vessel (the CFL and RBC-rich core) and the deviation of cell trajectories from flow streamlines in the vicinity of the bifurcation;

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and this phenomenon is responsible for the low capillary hematocrit distribution and in turn affects the flow resistance and distributions of oxygen and other metabolites in the microvasculature (Pries et al., 1986, 1996).

In addition to the extensive experimental investigations (Fenton et al., 1985; Pries et al., 1986, 1989, 1996; Schmid-Schonbein et al., 1980), hematocrit separation at microvascular bifurcations has also been studied with various numerical approaches. For example, Perkio and Keskinen (1983) had developed a mathematical model of the hematocrit reduction through bifurcation by considering the non-uniform RBC concentration in microvessels. Enden and Popel (1992) had applied a multiphase model to simulate the flow field in bifurcations, and the separating surface and plasma skimming were investigated. However, in these studies, blood was considered as a continuous fluid and RBCs were assumed to follow the flow streamlines exactly. For microcirculation situations, the RBC size is actually comparable to the characteristic length of the local flow or even the bifurcation dimension itself. Under such conditions, the trajectory of a cell can deviate dramatically from the imaginary streamlines, due to the strong hydrodynamic interactions with the cells and the microvessel walls around it. To account for the particulate nature of microscopic blood flows, El-Kareh and Secomb (2000) had modeled a RBC as a rigid spherical cap, and the flow field in the bifurcation region was approximated by the flow around a cylindrical post. On the other hand, Hyakutake et al. (2006, 2008) had simulated the flow and motions of RBCs and liposome-encapsulated hemoglobin particles through a microvascular bifurcation using the lattice Boltzmann method (LBM), with RBCs treated as rigid circular c effect, Barber et al. (2008) had employed a spring-network model and examined the cell trajectory and separation through bifurcations. A similar study by Xiong and Zhang (2012) had further considered the effect of the suspending viscosity, and found that a higher suspending viscosity can enhance the phase separation. Clearly, these single-particle simulations cannot include the effects of aggregation, hydrodynamic interaction between RBCs, and hematocrit. For this aspect, Hyakutake et al. (2010) had simulated the separation process of a suspension of RBCs and liposome-encapsulated hemoglobin particles at bifurcations. The results are relatively preliminary and detailed information, for example, for CFL structure, RBC trajectory, and hematocrit separation, is absent. Recently, Li et al. (2012) had conducted three-dimensional (3D) dissipative particle dynamics simulations of RBC flows through a diverging–converging microchannel, with RBCs represented as soft particles. In this study, the aggregation had not been considered, and the effects of flow rates and daughter branch sizes had been lumped together, since different flow rates in branches were achieved by varying the branch widths (Li et al., 2012). Also, the channel sizes utilized in these studies are relatively large (20 μm in Ref. Hyakutake et al., 2010 and 40 μm in Ref. Li et al., 2012); c as in smaller vessels (Barber et al., 2008; Pries et al., 1989).

In this study, we perform two-dimensional (2D) simulations of multiple RBC flows through a symmetric bifurcation model, using the immersed-boundary lattice-Boltzmann model (Zhang et al., 2007). Different cell deformability, aggregation, and feeding hematocrit values are employed, and their effects on CFL and hematocrit distributions in the daughter branches are examined. Our results and analysis suggest that the overall hematocrit separation is mainly related to the CFL structure, while the hydrodynamic interaction can dramatically shift the trajectory of an individual cell during the separation process. We also carefully examine the detailed flow and pressure fields in the bifurcation region during the RBC separation process, and the major driving force is the pressure difference for RBCs flowing with the plasma stream, but the shear stress on the cell back for those that have been pushed against the bifurcation corner wall. The results of hematocrit separation are also compared with the empirical relationship, and a reasonable agreement is observed. The information revealed in this study could be interesting for relevant biomedical studies, such as the viscous effect of suspending medium in blood substitute development (Yalcin

et al., 2011) and biochip design for blood cell separation (Kim et al., 2012; Xue et al., 2011).

Numerical models and methods

Red blood cell (RBC) model

A healthy, undeformed human RBC has a biconcave shape, which can be described by the following equation proposed by Evans and Fung (1972):

$$\bar{y} = 0.5(1 - \bar{x}^2)^{1/2}(c_0 + c_1\bar{x}^2 + c_2\bar{x}^4), \quad -1 \leq \bar{x} \leq 1, \quad (1)$$

where $c_0 = 0.207$, $c_1 = 2.002$, and $c_2 = -1.122$. The non-dimensional coordinates $(\bar{x}, \bar{y})$ are normalized by the typical radius of a human RBC 3.91 μm . The RBC membrane is highly deformable with a finite bending resistance, which becomes profound in regions of the cell membrane with large curvature. In our present 2D model, the neo-Hookean elastic component of membrane stress is expressed as

$$S_e = E_s \frac{\epsilon^3 - 1}{\epsilon^{3/2}}, \quad (2)$$

where E_s is the membrane elastic modulus and ϵ is the stretch ratio. The bending resistance can be incorporated by relating the change in membrane curvature to the membrane stress with (Pozrikidis, 2001)

$$S_b = E_b \frac{d}{dl}(\kappa - \kappa_0), \quad (3)$$

where E_b is the bending modulus, and κ and κ_0 are, respectively, the instantaneous and initial stress-free membrane curvatures; l is a measure of the arc length along the membrane surface. Membrane viscosity, which could be important in determining the cell deformation and motion, is not considered in this study for simplicity.

For normal RBCs, the viscosity of cytoplasm is 5 times of that of the suspending plasma. As the cell moves and deforms with the fluid flow, the fluid viscosity at a computational node has to be updated accordingly. Here we define a fluid node index d as the shortest distance to the membrane. If the node is close to two or more membrane segments, the index taken is that to the closest one. The index also has a sign, which is positive in the plasma region and negative in the cytoplasm region. A Heaviside function

$$\theta(d) = \begin{cases} 0, & d < -2\Delta x \\ \frac{1}{2} \left(1 + \frac{d}{2\Delta x} + \frac{1}{\pi} \sin \frac{\pi d}{2\Delta x} \right), & -2\Delta x \leq d \leq 2\Delta x \\ 1, & d > 2\Delta x \end{cases} \quad (4)$$

is utilized to obtain a smooth transition in viscosity across the membrane:

$$\mu(\mathbf{x}) = \mu_{in} + (\mu_{out} - \mu_{in})\theta[d(\mathbf{x})], \quad (5)$$

where μ_{in} and μ_{out} are, respectively, the viscosities in the cytoplasm and plasma regions.

To incorporate the aggregation effect, here we model the inter-cellular interaction energy ϕ as a Morse potential:

$$\phi(r) = D_e \left[e^{2\beta(r_0 - r)} - 2e^{\beta(r_0 - r)} \right], \quad (6)$$

where r is the distance between two adjacent membrane surfaces, r_0 and D_e are, respectively, the zero force separation and aggregation strength, and β is a scaling factor controlling the interaction decay behavior. The interaction force from such a potential is its negative derivative, i.e., $f(r) = -\partial\phi/\partial r$. This Morse potential was selected for its

simplicity and realistic physical description of the cell–cell interaction; however, it does not imply any particular underlying mechanism.

Lattice Boltzmann method (LBM) for fluid dynamics

In LBM, a fluid is modeled as pseudo-particles moving over a lattice domain at discrete time steps. The major variable in LBM is the density distribution $f_i(\mathbf{x}, t)$, indicating the particle amount moving with the i -th lattice velocity $\mathbf{c}_i$ at position $\mathbf{x}$ and time t . The time evolution of density distributions is governed by the so-called lattice Boltzmann equation

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = f_i(\mathbf{x}, t) + \Lambda_i(f), \quad (7)$$

which can be considered as a discrete version of the Boltzmann equation in classical statistical physics (Succi, 2001). Here Δt is the time step, and Λ_i is the collision operator incorporating the change in f_i due to the particle collisions. The collision operator is typically simplified with the single-time-relaxation approximation (Succi, 2001)

$$\Lambda_i(f) = -\frac{f_i(\mathbf{x}, t) - f_i^{eq}(\mathbf{x}, t)}{\tau}, \quad (8)$$

where τ is a relaxation parameter. The equilibrium distribution f_i^{eq} can be expressed as

$$f_i^{eq} = \rho \alpha_i \left(1 + \frac{\mathbf{u} \cdot \mathbf{c}_i}{c_s^2} + \frac{1}{2} \left(\frac{\mathbf{u} \cdot \mathbf{c}_i}{c_s^2} \right)^2 - \frac{u^2}{2c_s^2} \right). \quad (9)$$

Here $\rho = \sum_i f_i$ is the fluid density and $\mathbf{u} = \sum_i f_i \mathbf{c}_i / \rho$ is the fluid velocity. Other parameters, including the lattice sound speed c_s and weight factors α_i , depend on the lattice structure employed. Through the Chapman–Enskog expansion (Succi, 2001), one can recover the macroscopic continuity and momentum (Navier–Stokes) equations from the above-defined LBM microdynamics:

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) &= 0, \\ \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} &= -\frac{1}{\rho} \nabla P + \frac{\mu}{\rho} \nabla^2 \mathbf{u}, \end{aligned} \quad (10)$$

where $\mu = (2\tau - 1)c_s^2 \Delta t / 2\rho$ is the fluid viscosity and $P = c_s^2 \rho$ is the fluid pressure. For the D2Q9 (2D and nine-lattice velocity) lattice model utilized in our current study, the nine lattice velocities are

$$\begin{aligned} \mathbf{c}_0 &= [0, 0]; \\ \mathbf{c}_i &= \left[\cos \frac{(i-1)\pi}{2}, \sin \frac{(i-1)\pi}{2} \right] \frac{\Delta x}{\Delta t}, \quad i = 1-4; \\ \mathbf{c}_i &= \left[\cos \frac{(2i-9)\pi}{4}, \sin \frac{(2i-9)\pi}{4} \right] \frac{\Delta x}{\Delta t}, \quad i = 5-8, \end{aligned} \quad (11)$$

where Δx is the lattice grid spacing. The lattice weight factors for the D2Q9 model are $\alpha_0 = 4/9$, $\alpha_{1-4} = 1/9$, and $\alpha_{5-8} = 1/36$; and the lattice sound speed is $c_s = (1/\sqrt{3})\Delta x/\Delta t$. To simulate incompressible flows and to maintain the numerical stability, the Mach number in a LBM simulation must be low (Zhang, 2011a). For example, in our simulations in this study, the Mach number is $\sim 10^{-3}$, and thus the flow incompressibility is well enforced.

Immersed boundary method (IBM) for flow–cell interaction

In the 1970s, Peskin (1977) developed the novel immersed boundary method (IBM) to simulate flexible membranes in fluid flows. Since then, IBM has been widely employed to study various flow situations.

The membrane force $\mathbf{f}(\mathbf{x}_m)$ at a membrane node $\mathbf{x}_m$ induced by membrane deformation is distributed to the nearby fluid nodes $\mathbf{x}_f$ by

$$\mathbf{F}(\mathbf{x}_f) = D(\mathbf{x}_f - \mathbf{x}_m) \mathbf{f}(\mathbf{x}_m), \quad (12)$$

through a discrete delta function $D(\mathbf{x})$, which is chosen to approximate the properties of the Dirac delta function (Peskin, 1977). In our 2D system, $D(\mathbf{x})$ is given as

$$\begin{aligned} D(\mathbf{x}) &= \frac{1}{4(\Delta x)^2} \left(1 + \cos \frac{\pi x}{2\Delta x} \right) \left(1 + \cos \frac{\pi y}{2\Delta x} \right), \\ &\quad |x| \leq 2\Delta x \quad \text{and} \quad |y| \leq 2\Delta x \\ D(\mathbf{x}) &= 0, \quad \text{otherwise.} \end{aligned} \quad (13)$$

The membrane velocity $\mathbf{u}(\mathbf{x}_m)$ can be obtained in a similar way according to the local flow field:

$$\mathbf{u}(\mathbf{x}_m) = \sum_f D(\mathbf{x}_f - \mathbf{x}_m) \mathbf{u}(\mathbf{x}_f). \quad (14)$$

The no-slip requirement between the fluid and the membrane is automatically satisfied, since the membrane points are moving exactly at the same velocity as the local flow field, within negligible numerical errors.

In addition, the fluid viscosity at each lattice node in the computational domain has been updated simultaneously with RBC motion and deformation (Zhang et al., 2007). This immersed boundary lattice Boltzmann model (IB-LBM) has been validated in previous publications (Zhang, 2011b; Zhang et al., 2007) and utilized in several microscopic blood flow studies (Vahidkhah and Fatourae, 2012; Yin and Zhang, 2012a).

Simulation parameters, setups, and processes

Simulation parameters and cases

In our simulations, the membrane of a RBC is represented by 100 elastic elements. The simulation domains are discretized into uniform square meshes with a lattice resolution of $\Delta x = 0.1955 \mu\text{m}$, and the D2Q9 lattice structure is utilized. The RBC membrane elasticity and fluid viscosity values are selected from experimental measurements. The RBC aggregation parameters are not available and their values are chosen based on our previous studies (Zhang et al., 2007, 2008, 2009). These parameters are listed in Table 1 with references. To examine the individual effects from cell deformability, aggregation, and

Table 1
Simulation parameters used in this work with references.

Property	Symbol	Value
Lattice grid unit	Δx	0.1955 μm
Time step	Δt	2.66×10^{-8} s
Fluid density	ρ	1000 kg/m ³ (Skalak and Chien, 1987)
Plasma viscosity	μ_p	1.2 cP (Skalak and Chien, 1987)
Cytoplasm viscosity	μ_c	6.0 cP (Vaughan and Hochmuth, 2006)
Mean flow velocity	U_0	1.0 mm/s (Barber et al., 2008)
Membrane elastic modulus	E_s	1.5×10^{-5} – 1.5×10^{-4} N/m (Lim et al., 2004; Tan et al., 2010)
Membrane bending modulus	E_b	5.0×10^{-19} – 5.0×10^{-18} Nm (Lim et al., 2004; Tan et al., 2010)
Aggregation strength	D_e	0 – 5.2×10^{-7} $\mu\text{J}/\mu\text{m}^2$ (Yin and Zhang, 2012a)
Scaling factor	β	$3.84 \mu\text{m}^{-1}$ (Yin and Zhang, 2012a)
Zero force distance	r_0	0.49 μm (Yin and Zhang, 2012a)

hematocrit on RBC separation, four cases are considered in this current study as follows:

- Case C: This system consists of RBCs with normal deformability ($E_s = 1.5 \times 10^{-5}$ N/m and $E_b = 5.0 \times 10^{-19}$ Nm) (Lim et al., 2004; Tan et al., 2010) with no aggregation considered. The tube hematocrit (obtained by dividing the total RBC area by the rectangular domain area of the channel flow simulations, which is specified below in section Channel flow simulation) in the mother vessel is $H_{ct} = 32.32\%$. This system is taken as the control system and serves as the comparison basis with other cases, which differentiate from this control case, respectively, by only one aspect.
- Case R: The cell deformability is reduced by increasing the membrane elastic and bending moduli 10 folds (Xiong and Zhang, 2012) (i.e., $E_s = 1.5 \times 10^{-4}$ N/m and $E_b = 5.0 \times 10^{-18}$ Nm). All other parameters remain the same as in Case C.
- Case A: The aggregation effect is introduced via Eq. (6) with the aggregation strength $D_e = 5.2 \times 10^{-7}$ $\mu\text{J}/\mu\text{m}^2$ (Yin and Zhang, 2012a; Zhang et al., 2008). All other parameters remain the same as in Case C.
- Case D: A lower tube hematocrit of $H_{ct} = 26.26\%$ (diluted condition) is obtained by reducing the RBC numbers in the simulations (see details in the next section). All other parameters remain the same as in Case C.

Channel flow simulation

To have a better representation of the RBC deformation and distribution in the feeding flow toward the bifurcation, simulations of RBC flows in a straight channel are conducted for each case. The straight channel has a length of $L = 349\Delta x = 68.23 \mu\text{m}$ and a width of $H = 51\Delta x = 9.97 \mu\text{m}$, the same width as the mother vessel in the bifurcation model to be described below. At the beginning of the simulation, 16 (for Cases C, R, and A) or 13 (for Case D) undeformed RBCs are randomly placed in the channel, resulting in a tube hematocrit of $H_{ct} = 32.32\%$ or $H_{ct} = 26.26\%$, which is in the normal hematocrit range in the microvasculature. Periodic boundaries are applied to the two channel ends, and so the simulations can be considered as for RBC flows in an infinitely long channel. A body force is applied along the channel to drive the RBC flow. For such a suspension flow with deformable RBCs, the resulting flow rate cannot be obtained from the applied force in advance. Therefore, we start with the body force calculated from the Poiseuille law for a pure plasma flow with a mean velocity of 1 mm/s, and then in the later part of the simulation, dynamically adjust the driving force by comparing the simultaneous

mean velocity with the desired one 1 mm/s, till a quasi-steady state (Zhang et al., 2009) is established.

Bifurcation flow simulation

Similar to previous studies (Hyakutake et al., 2008, 2010; Xiong and Zhang, 2012), a symmetric bifurcation model with a 90° branching angle is constructed as shown in Fig. 1. The mother vessel is $H_0 = 51\Delta x = 9.97 \mu\text{m}$ wide and $1000\Delta x = 195.5 \mu\text{m}$ long. The daughter branch width is $H_1 = H_2 = 40.48\Delta x = 7.91 \mu\text{m}$, according to the empirical cubic dependence (Mayrovitz and Roy, 1983). The corners where two vessel walls intersect are rounded up by circular arcs with a radius of $3 \mu\text{m}$. A relatively large domain ($2100\Delta x \times 2400\Delta x$, only part of it displayed in Fig. 1) is necessary so that we can observe the entire separation process before any cells moving out of the boundaries; however, the LBM calculation is limited on the fluid lattice nodes only inside the microvessels. The no-slip fluid-wall boundary condition is implemented by using an improved bounce-back scheme recently developed for LBM (Yin and Zhang, 2012b). Parabolic velocity profiles are imposed at the two branch exits, according to the required flow rates Q_1 at the upper exit and Q_2 at the lower exit, while the flow rate in the mother vessel $Q_0 = Q_1 + Q_2$ is maintained constant at that for a mean velocity of 1 mm/s. At the left inlet to the mother vessel, an extrapolation boundary condition (Zhang, 2011a) is adopted by assuming linear variations in fluid properties along the mother vessel.

To populate the mother vessel with deformed RBCs of the particular CFL-core distribution, a snapshot is taken from the channel flow simulation at the quasi-steady state. Each deformed cell in the snapshot is then placed to the mother vessel twice, with the following translations:

$$(x_1, y_1) = (x + L_0, y), \quad \text{and} \quad (x_2, y_2) = (x + L_0 + L, y), \quad (15)$$

where (x, y) is the original position of a RBC membrane node in the channel simulation snapshot, and (x_1, y_1) and (x_2, y_2) are the locations in the mother vessel for the bifurcation simulation (Fig. 1). An offset $L_0 = 100\Delta x$ in the x -direction is considered to leave a segment of pure plasma between the RBCs and the left inlet, so that the extrapolation boundary condition at the inlet is valid and the calculation is stable. By duplicating the RBCs from the channel to the mother vessel, we have doubled the RBC numbers in our bifurcation simulations: 32 RBCs for Cases C, R, and A, and 26 RBCs for Case D. This larger sample space is helpful to reduce the statistical noise for our analysis of RBC separation. Moreover, similar to those in our previous study (Xiong and Zhang, 2012), a repulsion force is introduced to a membrane node if it is too close to the vessel wall, and a restoring force is

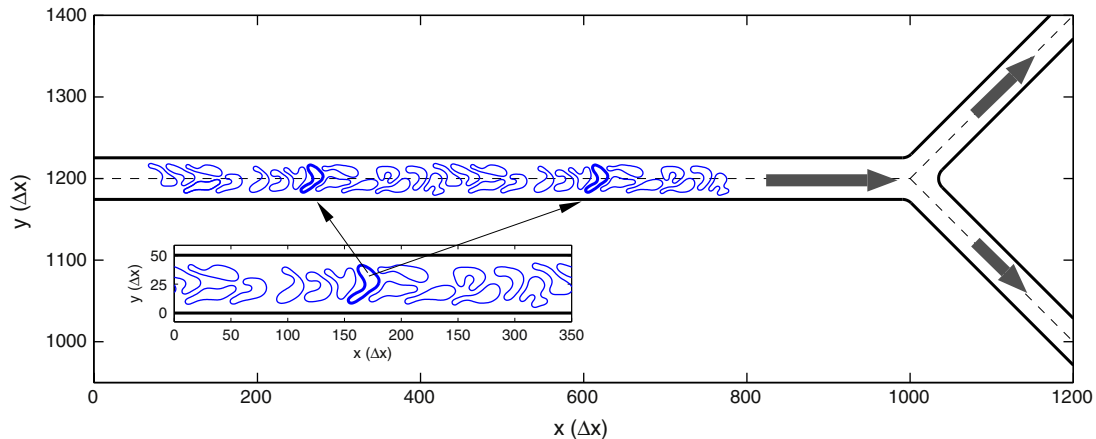


Fig. 1. The symmetric bifurcation model used in this study. The initial distribution of deformed RBCs in the mother vessel is obtained from a snapshot of the corresponding channel flow simulation (inset). The three thick arrows indicate the flow directions at the bifurcation. See text for detailed descriptions of the boundary conditions imposed.

adopted to maintain a constant cell area. Detailed descriptions on these treatments can be found in our previous publication (Xiong and Zhang, 2012). Please note that, although the bifurcation simulations here start with deformed cell shapes from the corresponding channel simulations, the membrane force is still calculated with respect to the original biconcave stress-free RBC shape defined in Eq. (1).

Results and discussions

Cell free layer (CFL) thickness

For each hematocrit level (32.32% and 26.26%), two initial configurations are designed with undeformed RBCs randomly and relatively evenly placed in the channel. Starting with such an initial configuration, the channel simulation is run, and three consecutive snapshots, at an interval of 2×10^5 time steps (5.31 ms) apart, are taken when the quasi-steady state has been established. For each snapshot, the Heaviside function value θ (see Eq. (4)) in the channel is averaged along the channel direction:

$$\bar{\theta}(y) = \frac{1}{L} \int_0^L \theta(x', y) dx', \quad (16)$$

where $L = 349\Delta x$ is the channel length. The profile of $\bar{\theta}(y)$ is then integrated from the bottom channel wall upward and normalized with the total integrated value over the entire channel width:

$$f(y) = \int_0^y \bar{\theta}(y') dy' / \int_0^H \bar{\theta}(y') dy', \quad (17)$$

where $H = 51\Delta x$ is the channel width. This function $f(y)$ represents the fraction of total RBC volume below a certain transverse location y , and we have $f(0) = 0$ and $f(H) = 1$ (Fig. 2). We define the CFL boundary as the location where there is only 2.5% of the total RBC volume from the channel wall to this location, i.e., y_l from $f(y_l) = 0.025$ for the CFL near the lower wall, and y_u from $f(y_u) = 0.975$ for the CFL

near the upper wall. By such a definition, we have 95% of the RBC volume included in the central core region. The CFL thickness for the flow is then calculated as the average of those measured near the lower and upper walls:

$$\delta = \frac{y_l + (H - y_u)}{2}. \quad (18)$$

For Case C, the six δ -values are 1.99, 1.98, and 1.97 μm from one initial configuration, and 1.90, 1.88, and 1.85 μm from another initial configuration. The mean value of them is 1.93 μm , and the standard deviation is 0.059 μm . The consistency of these values indicates that our method can provide a reasonable measure of the CFL structure for further quantitative analysis. This CFL value is similar to that reported in a 3D dissipative particle dynamics simulation, but larger than those measured in experiments (Fedosov et al., 2010). One possible explanation of this difference could be the not fully-developed flow structure in the in vivo experiments (Fedosov et al., 2010; Zhang, 2011b). Furthermore, the definition of the CFL boundary is sort of arbitrary in our calculation. If we include 98% or even 99% of the RBC volume in the RBC-core region, the CFL thickness from Fig. 2 will decrease from 1.93 to 1.73 and 1.27 μm , respectively.

RBC separation and trajectory

Previous studies of RBC trajectory through microvascular bifurcations had considered only single cells, and the cell was initially assumed as undeformed (Barber et al., 2008; Xiong and Zhang, 2012). The particular cell deformation and distribution in the feeding flow as well as the hydrodynamic interaction among cells cannot be incorporated. We now look at the cell separation and trajectories for our multiple RBC flow as the cells pass the bifurcation region. To illustrate such a dynamic and continuous process, Fig. 3 displays several representative snapshots during the separation process for Case C with a flow rate ratio $Q_1/Q_0 = 0.7$. The cells flowing into the upper branch (high flow rate) are plotted in red lines; and those entering the lower

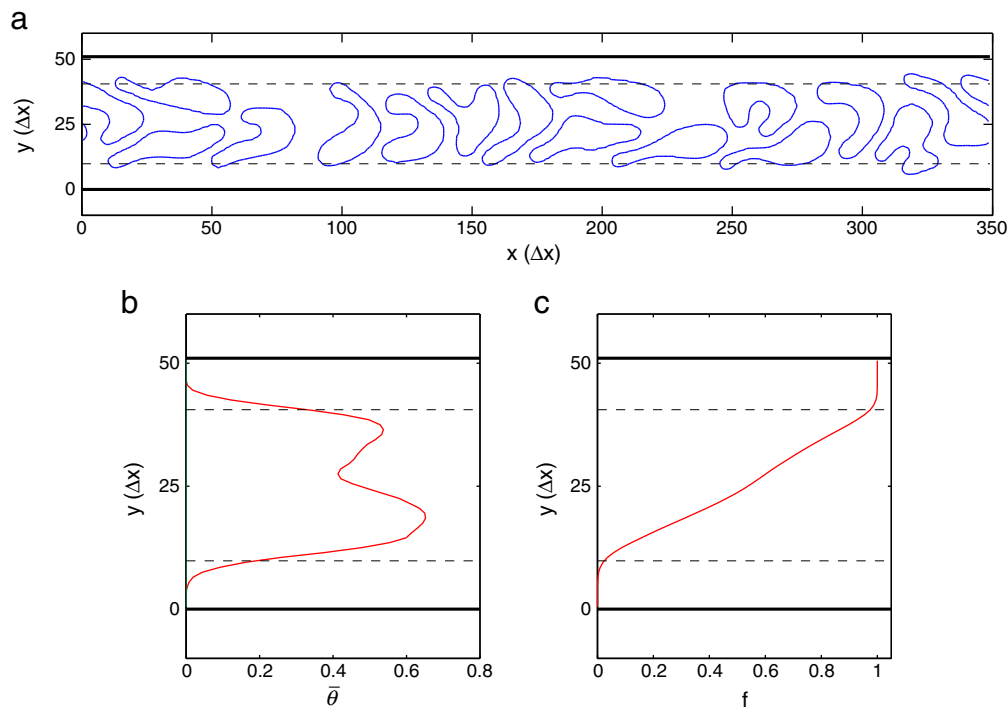


Fig. 2. The calculation of CFL thickness: (a) a snapshot of the cell deformation and distribution in the channel at the quasi-steady state, (b) the profile of averaged Heaviside function value, and (c) the profile of the integrated RBC volume fraction from the lower wall. The dashed lines indicate the CFL locations y_l and y_u according the method described in the text.

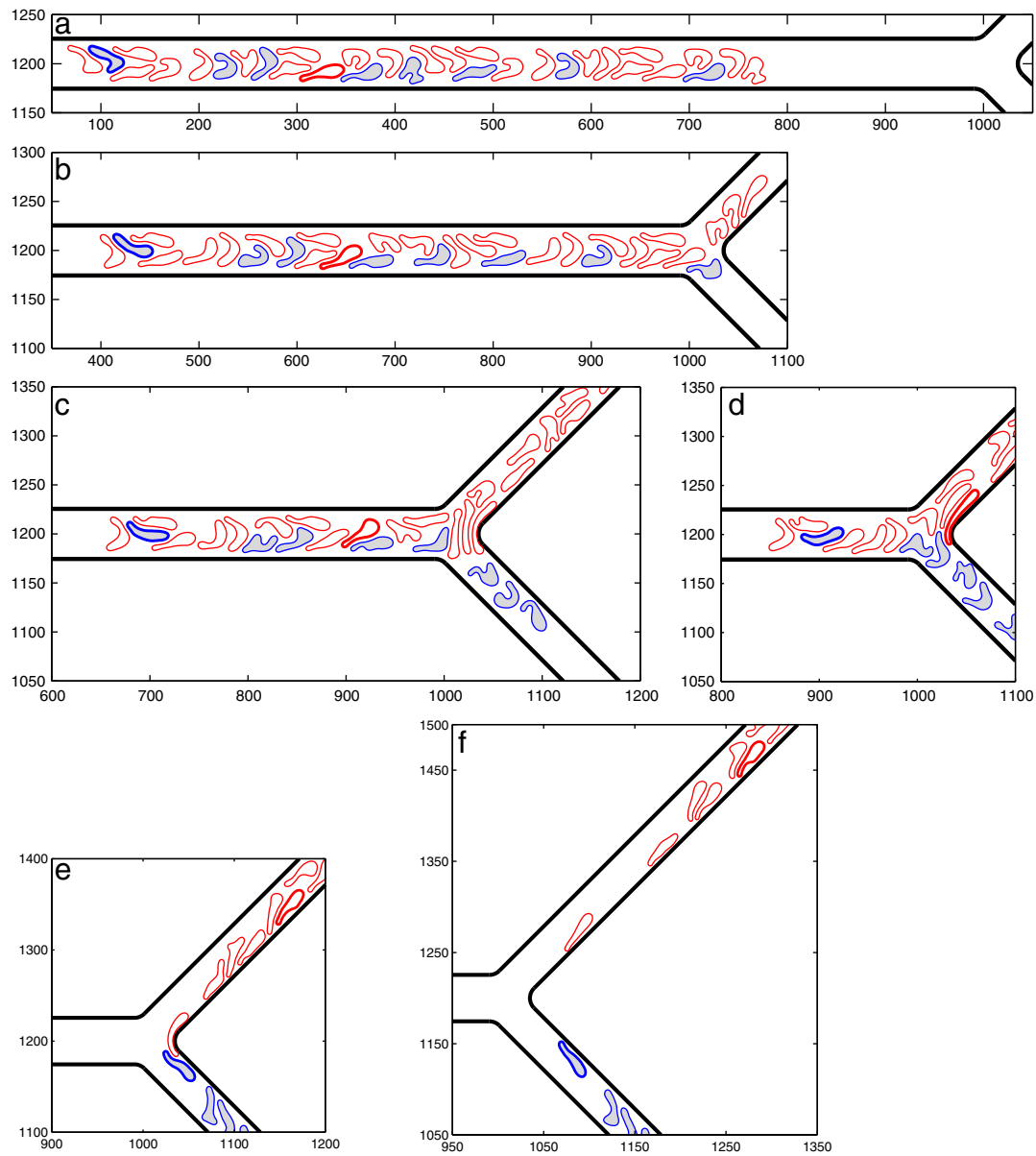


Fig. 3. Representative snapshots showing the separation process for Case C with flow rate ratio $Q_1/Q_0 = 0.7$. Cells plotted in white wide red edges are going to the upper branch, while cells in gray with blue edges to the lower branch. The two particular cells A (gray) and B (white) mentioned in text are displayed with thick edges for clarity.

branch (low flow rate) are displayed in blue lines. For this simulation, we have 24 RBCs moving to the upper branch, and 8 going to the lower branch. The tube hematocrits calculated from these RBC numbers and the flow rates are 34.63% for the upper branch ($Q_1 = 0.7Q_0$) and 26.93% for the lower branch ($Q_2 = 0.3Q_0$). Recall the tube hematocrit in the feeding flow from the mother vessel is 32.32%, we see that the hematocrit has increased in the high flow rate branch but decreased in the low flow rate branch. This is right the phase separation phenomenon observed in experiments.

The cell motion in the bifurcation region of such a dense RBC flux is, unsurprisingly, a complicated process. In Fig. 4 we plot the pressure (background colors) and velocity (arrows) distributions in the bifurcation region, for the same time instances shown in Fig. 3. Since in fluid mechanics only the pressure gradient matters in determining the flow behaviors, we adjust the absolute pressure values to have a same mean pressure at location $x = 900$ for all time instances. Also the pressure distributions are plotted using a same color scale for the convenience of our comparison. Before the RBCs arrive at the

bifurcation region, the flow is not disturbed and we see a smooth separating flow there, with the velocity distributions in the mother and daughter microvessels approximately parabolic except the bifurcation center region (Fig. 4a). Also the pressure decreases gradually along the flow directions into the two branches, a little faster in the upper branch than the lower branch, due to the larger flow rate in the upper vessel. During the separation process, the RBCs in the bifurcation region increase the flow resistance significantly, and larger pressure drops are required to overcome such a stronger resistance, as indicated by the low pressures in the daughter vessels (Figs. 4b–d). Meanwhile, when there are cells that have been pushed against the bifurcation corner (Fig. 4c), the available flow passage is greatly reduced, and the separating point moves inward (e.g., compare Figs. 4c to b). At such particular conditions, which have also been noticed in in vivo observations (Baskurt and Meiselman, 2003), a high pressure is also built up between those stacked cells and the bifurcation corner (Fig. 4b). When the separation finishes (Figs. 4e and f), the flow recovers to the non-cell free flow condition as in Fig. 4a.

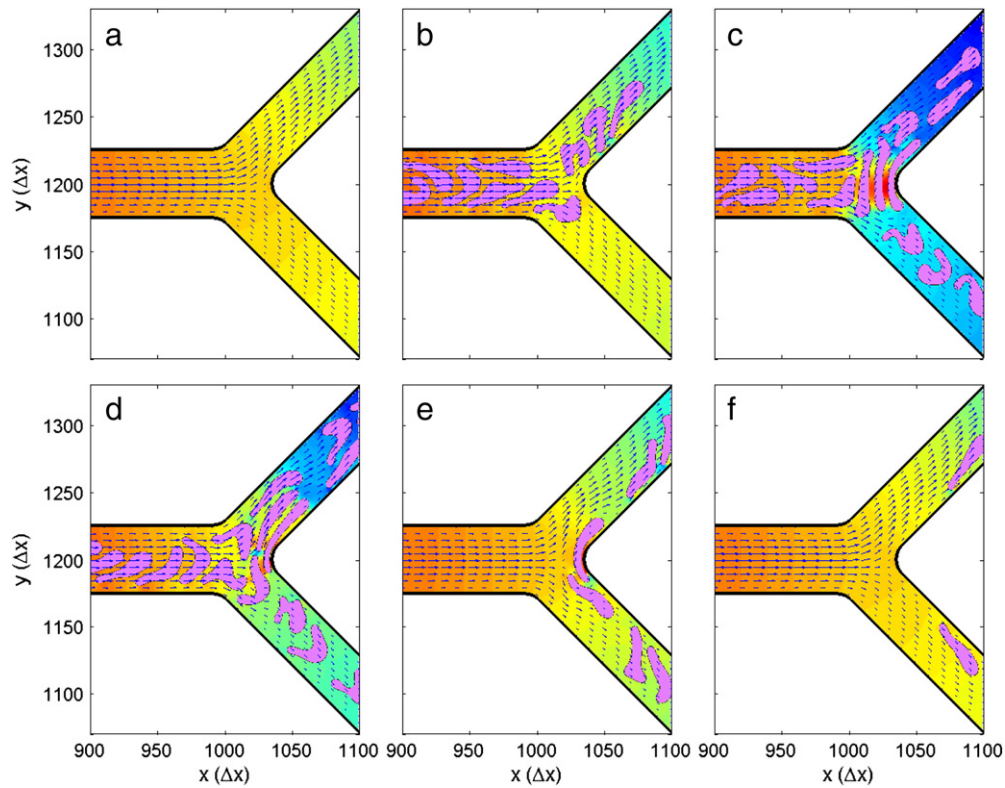


Fig. 4. Pressure (background colors) and velocity (arrows) distributions in the bifurcation region at the same time instances plotted in Fig. 3. The mean pressure values in the mother vessel at $x = 900$ are adjusted to be the same, and the adjusted pressure are displayed using a same color scale for a direct comparison.

In principle, the process we study is a flow–cell interaction system, and the cell motion is driven by the hydrodynamic forces, including pressure and shear stress, exerted on the cell membrane. For RBCs flowing with the plasma streams, we typically see a pressure difference between the front and rear ends of the cells, while the flow velocity is relatively uniform (Figs. 4b and d). This means that, for such flowing cells, the major driving force through the bifurcation is the pressure force. On the other hand, for those cells that have been pushed against the separation corner wall (see the corner cells in Figs. 4c and e), the relative slow sliding along the corner wall into a branch is mainly due to the shear stress applied on the back side of

the cell. In addition, similar to that we have noticed in previous studies (Xiong and Zhang, 2010; Yin and Zhang, 2012a), when a cell is moving close to the vessel wall (see cells in the upper branch in Figs. 4b, e, and f), a local high-pressure region is developed in front of the cell end near the vessel surface, and a low-pressure area is observed at the back of the cell end. In addition to cause shear stress variations on the vessel wall (Xiong and Zhang, 2010; Yin and Zhang, 2012a), this particular local pressure field could also be responsible for the cell free layer development. This hypothesis will be further investigated in a future study.

In Fig. 5, we plot the trajectories of RBC centers during the separation process. For clarity, we plot the cells entering the upper branch in

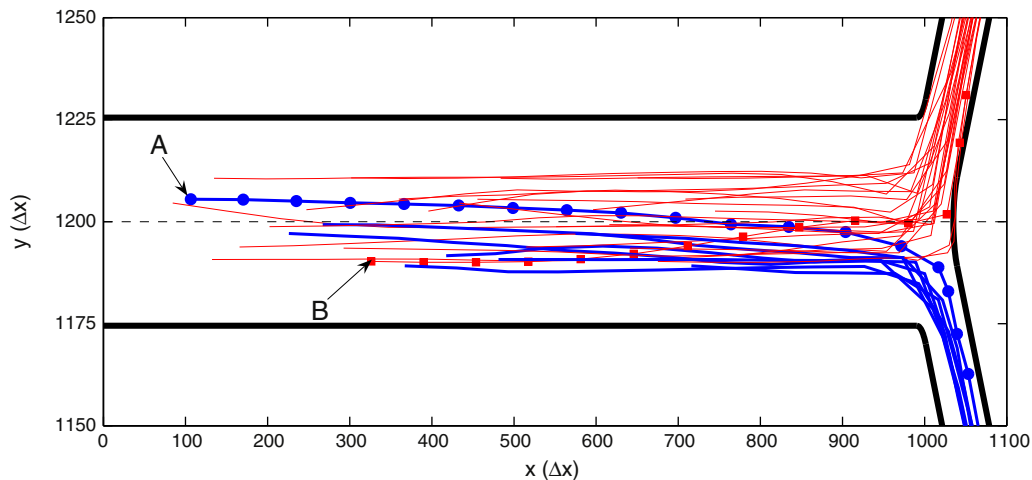


Fig. 5. RBC trajectories over the separation process for Case C with flow rate ratio $Q_1/Q_0 = 0.7$. Trajectories plotted in thin and red lines are for cells going to the upper branch, while thick and blue lines for those to the lower branch. The trajectories for cells A and B (with thick edges in Fig. 3) are also highlighted with thick lines and symbols for clarity.

white with red edges, and those going to the lower branch after the separation completed in gray with blue edges. Previous studies had indicated that a separating surface (Enden and Popel, 1992), or a separating location in 2D systems (Barber et al., 2008; Xiong and Zhang, 2012), exists, which divides the cross-section of the mother vessel into two regions: RBCs centering in one region will flow into one branch, and others in another region will enter the other branch. However, with these two group trajectories (thin, red lines and thick, blue lines) intersecting with each other, we cannot find such a separating location in Fig. 5. More specifically, Cell A (in gray with a blue thick edge in Fig. 3) originally locates above the centerline of the mother vessel, and it enters into the lower branch with a lower flow rate. On the other side, Cell B (in white with a red thick edge in Fig. 3) is very close to the lower wall in the mother vessel at the beginning of the simulation; however, it eventually moves to the upper branch (Figs. 3 and 5). Clearly, for such a particulate flow with the cell size comparable to that of the system, the simultaneous hydrodynamic interaction, in addition to the initial position of the cell in the mother vessel and the flow rate ratios in branches, is also playing an important role in determining the cell destination. Nevertheless, we still see that most of the cells going to the lower branch indeed initially locate close to the lower wall of the mother vessel, and vice versa. The concept of a separating surface or separation location could be useful for a statistical description of the RBC separation, however, it cannot be used to predict the destination of a particular cell, especially for concentrated RBC flows in small vessels.

Hematocrit separation

The six snapshots taken for CFL calculations of each case (section Cell free layer (CFL) thickness) are also used to initialize the RBC configuration in the mother vessel for the bifurcation simulation, as described in section Bifurcation flow simulation. For each initial configuration from a snapshot, different flow rate ratios, from $Q_1/Q_0 = 0.05$ to 0.95 with an interval of 0.05 , are applied at the exits, and the RBC numbers entering the upper branch are recorded. Furthermore, for a particular initial configuration, if we have N_u RBCs flowing into the upper branch with a flow rate Q_1 , considering the symmetric bifurcation model here, we can say that there will be $N - N_u$ (N is the total RBC number in the bifurcation simulation) RBCs going to the upper branch when we flip the initial RBC configuration about the mother vessel centerline and set the upper branch flow rate as $Q_0 - Q_1$. Therefore, in total, we have $12 N_u$ values for each flow rate ratio Q_1/Q_0 for each case defined in section Simulation parameters and cases.

For Case C, Fig. 6 shows the RBC flux ratio N_u/N changing with the flow rate ratio Q_1/Q_0 between flows in the upper branch and the mother vessel, with the circles for the mean values and the error bars for the standard deviations. Also displayed there as a dashed line is the empirical relationship by Pries and Secomb (2005) with the mother vessel diameter of $9.97 \mu\text{m}$ and the feeding hematocrit of 32.32% , and the straight, dot-dashed line $N_u/N = Q_1/Q_0$ indicates the situation with no phase separation (e.g., small particles uniformly distributed in a large bifurcated system). Our simulated results for the current Case C lie between them, meaning that the phase separation has been reproduced correctly, but less significant than that from the empirical relationship. Several factors might be responsible for this discrepancy. First, the endothelial glycocalyx layer has not been considered in the empirical relationship. If we assume a $0.5\text{--}1 \mu\text{m}$ thick glycocalyx layer on the vessel surface, the $9.97 \mu\text{m}$ wide mother vessel in our bifurcation simulation actually should correspond to a microvessel with a diameter of $10.97\text{--}11.97 \mu\text{m}$ in the empirical relationship, and a wider mother vessel is associated with a less significant phase separation in hematocrit. Another factor is that, for the current Case C, we do not consider the aggregation among RBCs, which can enhance the phase separation at the bifurcation, as we will see in section Cell deformability, aggregation, and hematocrit

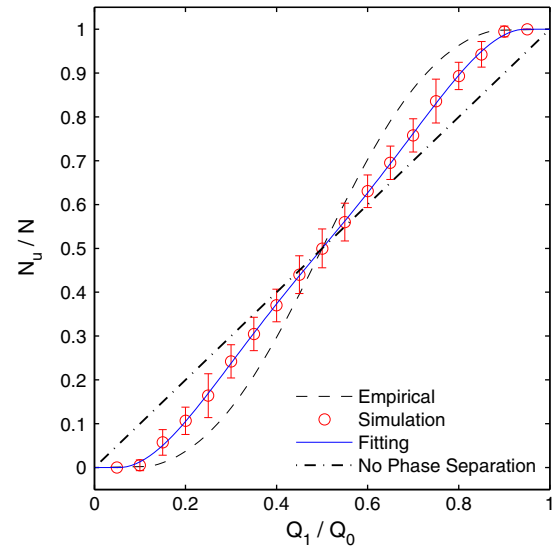


Fig. 6. Simulated RBC flux ratio vs. the imposed flow rate ratio for Case C (circles and error bars). The dashed curve from the empirical relationship with feeding hematocrit 32.32% and mother vessel diameter $9.97 \mu\text{m}$ and the dot-dashed straight line for no-phase-separation situation are also displayed for comparison.

effects. At last, the 2D model in our study is only a simple approximation compared to the real 3D systems, and we should not expect a perfect agreement.

For the convenience of our further quantitative analysis of the hematocrit separation under different conditions, we fit the mean RBC flux ratio values by the following function with two free parameters α and β :

$$\frac{N_u}{N} = \begin{cases} 0, & \frac{Q_1}{Q_0} - \frac{1}{2} \leq -\alpha; \\ \frac{1}{2} + \beta \sin \left[\frac{\pi}{2\alpha} \left(\frac{Q_1}{Q_0} - \frac{1}{2} \right) \right] + (1-\beta) \left\{ \frac{1}{\alpha} \left(\frac{Q_1}{Q_0} - \frac{1}{2} \right) \right. \\ \quad \left. + \frac{1}{\pi} \sin \left[\frac{\pi}{\alpha} \left(\frac{Q_1}{Q_0} - \frac{1}{2} \right) \right] \right\}, & \alpha < \frac{Q_1}{Q_0} - \frac{1}{2} < \alpha; \\ 1, & \frac{Q_1}{Q_0} - \frac{1}{2} \geq \alpha. \end{cases} \quad (19)$$

This particular piecewise function is designed to satisfy several criteria, including the symmetric shape about point $(Q_1/Q_0 = 1/2, N_u/N = 1/2)$, smooth transitions at $Q_1/Q_0 - 1/2 = \pm \alpha$, and $N_u/N = 0$ when $Q_1/Q_0 - 1/2 \leq -\alpha$ and $N_u/N = 1$ when $Q_1/Q_0 - 1/2 \geq \alpha$, to account for the fact that all cells will go to the high flow rate branch and only pure plasma is flowing into the low flow rate branch (plasma skimming) when the flow rate difference is large enough. The fitting curve is also shown in Fig. 6 as the solid curve. We then define the critical flow rate ratio $(Q_1/Q_0)_{0.95}$, at which 95% of the feeding RBCs will enter this branch. Compared with the 100%-up flow rate ratio $\alpha + 1/2$, $(Q_1/Q_0)_{0.95}$ has a better representation of the fitting curve, including its overall inclination and local curvature. In Fig. 6, this critical flow rate ratio is 0.851 for the simulated results according to the fitting curve, 0.782 for the empirical relationship, and 0.95 for the no-phase-separation situation straight line. This parameter can then serve as a quantitative indicator of the phase separation phenomenon: the smaller the critical flow rate ratio, the stronger the phase separation.

Cell deformability, aggregation, and hematocrit effects

The above described channel and bifurcation simulations and relevant CFL and phase separation analysis are performed for all the four

Table 2
Calculated CFL thickness δ and critical flow rate ratio $(Q_1/Q_0)_{0.95}$ for each case.

Case	δ (μm)	$(Q_1/Q_0)_{0.95}$
C	1.93	0.851
R	1.38	0.882
A	2.12	0.807
D	2.16	0.787

cases described in section [Simulation parameters and cases](#), and the results of CFL thickness and critical flow rate ratios are summarized in [Table 2](#). The smallest CFL thickness is observed in Case R, since the less deformable RBCs can inhibit the RBC migration as well as prevent the establishment of a more concentrated RBC-rich core. On the other hand, the aggregation in Case A attracts RBCs together to generate a high-density RBC core in the central region, while a reduced hematocrit in Case D allows more space for the cell migration and the CFL. Therefore, the CFL thickness has increased in both cases compared to the control system Case C. This trend is consistent with those reported in previous studies ([Fedosov et al., 2010](#); [Ong et al., 2010](#); [Yalcin et al., 2011](#); [Zhang, 2011b](#); [Zhang et al., 2009](#)). The smaller thickness ($\sim 2 \mu\text{m}$) compared with experimental results ($\sim 1.5 \mu\text{m}$) ([Maeda et al., 1996](#)) could be due to the undeveloped CFL structure in the *in vivo* measurements and other issues, as explained in section [Cell free layer \(CFL\) thickness](#).

Since the non-uniform cell distribution in the mother vessel is one of the major factors causing the hematocrit phase separation at a bifurcation, it is not surprising to note the strong correlation between the CFL thickness and the critical flow rate ratio in [Table 2](#). The RBC aggregation and the reduced feeding hematocrit increase the CFL thickness, and this stronger heterogeneity in RBC distribution in the feeding flow enhances the hematocrit phase separation, resulting in a smaller critical flow rate ratio. The enhanced phase separation with a reduced hematocrit has also been observed in experiments ([Pries et al., 1989](#)). On the other hand, the decreased cell deformability has a counter effect on the CFL thickness as well as the phase separation, i.e., a thinner CFL and less profound phase separation. To estimate the possible effect of the glycocalyx layer on the empirical relationship by Pries and Secomb for phase separation, we also calculate the critical flow rate ratios from the empirical relationship using a feeding hematocrit of 32.32% and different mother vessel diameters, and the critical flow rate ratios obtained are: 0.782 for $9.97 \mu\text{m}$ (same as that in our model, no glycocalyx layer considered), 0.795 for $10.97 \mu\text{m}$ (by assuming a $0.5 \mu\text{m}$ thick glycocalyx layer), and 0.807 for $11.97 \mu\text{m}$ (by assuming a $1 \mu\text{m}$ thick glycocalyx layer). We see the critical flow rate ratio with a $11.97 \mu\text{m}$ wide mother vessel is actually very close to our simulated value for Case A.

Summary

We have simulated multiple RBC flows through a 2D symmetric bifurcation model with different cell deformability, aggregation, and feeding hematocrit. The hematocrit phase separation has been reproduced in our simulations, and the overall separation performance is mainly related to the non-uniform RBC distribution in the feeding flow: the thicker the CFL in the mother vessel, the more profound the phase separation at the bifurcation. In addition to the flow rate ratios to the daughter branches, the local hydrodynamic interaction around a RBC also plays a non-negligible role on its motion, trajectory, and destination during the separation. The driving force for the cells passing through the bifurcation region into a daughter branch is the pressure difference force for flowing RBCs with the plasma stream and the shear stress on the cell back for a RBC that has been pressed against the bifurcation corner wall. The separating surface or separating location reported in previous studies, where blood was treated as a continuum fluid or only single, isolated RBCs were simulated, cannot be used to describe individual cell motions in multiple RBC flows. The calculated

CFL and phase separation characters are in reasonable agreement with experimental results. This information is helpful in understanding the complex phase separation process, and could be useful for relevant biomedical applications, such as blood substitute development ([Hyakutake et al., 2008](#); [Vazquez et al., 2009](#)) and blood analysis ([Li et al., 2012](#)). We are aware of the inherent limitations of our 2D models, and the symmetric bifurcation model is relatively simple. More realistic simulations with 3D cell and microvessel models, continuous RBC fluxes, and even multiple bifurcation levels could be considered to obtain more reliable information. Also extensive studies would be desirable to examine the particular effects of vessel size, branching angle, flow rate, and suspending viscosity on the hematocrit separation, flow resistance, and gas transport ([Vazquez et al., 2009](#); [Xiong and Zhang, 2012](#); [Zhang, 2011b](#)).

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